

Martingales — Exam (set 3)

Name: _____

Date: _____

Duration: 3 hours. You may use any result from the course by citing it precisely. The questions can often be solved independently. Marks out of 100 (indicative). *Answer in the space provided; the rigour of your justifications matters.*

Problem I — Fixation Probability and Accumulated Genetic Variance in a Boundary-Slowed Genetic Drift Model (19 pts)

(indicative time: 28 min)

Let $N \in \mathbb{N}$, $N \geq 2$, be fixed. Consider a Markov chain $(X_n)_{n \in \mathbb{N}}$ taking values in the finite set $E = \{0, 1, \dots, N\}$, modelling the frequency of an allele in a population of size N (model of *genetic drift slowed at the boundaries*). The two states 0 and N are *absorbing*. From an *interior* state $k \in \{1, \dots, N-1\}$, the chain evolves as follows: setting

$$a_k = \frac{2k(N-k)}{N^2} \in (0, \frac{1}{2}],$$

we have

$$\mathbb{P}(X_{n+1} = k+1 \mid X_n = k) = a_k, \quad \mathbb{P}(X_{n+1} = k-1 \mid X_n = k) = a_k, \quad \mathbb{P}(X_{n+1} = k \mid X_n = k) = 1 - 2a_k.$$

We denote by $\mathcal{F}_n = \sigma(X_0, \dots, X_n)$ the natural filtration, and we start from a deterministic state $X_0 = x \in \{1, \dots, N-1\}$. It is admitted that the chain is well defined and that a_k is chosen exactly so that $a_k \leq \frac{1}{2}$ for all k (with equality possible when N is even, at $k = N/2$). We introduce the absorption time

$$\tau = \inf\{n \in \mathbb{N} : X_n \in \{0, N\}\}.$$

Notation. For a function $h : E \rightarrow \mathbb{R}$ and $k \in \{1, \dots, N-1\}$, we call *harmonicity condition at k* the equality

$$\mathbb{E}[h(X_{n+1}) \mid X_n = k] = h(k).$$

Question 1.

- (a) Show that τ is a stopping time for $(\mathcal{F}_n)_{n \in \mathbb{N}}$, and that $\mathbb{P}(\tau < \infty) = 1$.

(b) Show that the process $M_n := X_n$ is a bounded martingale with respect to $(\mathcal{F}_n)$.

Question 2. By applying the optional stopping theorem to (M_n) , compute, for every $x \in \{1, \dots, N-1\}$, the *fixation* probability

$$u(x) = \mathbb{P}(X_\tau = N).$$

The use of the optional stopping theorem should be justified precisely.

Question 3. Set

$$v_k = 4 \frac{k(N-k)}{N^2} \quad (k \in E), \quad A_n = \sum_{i=0}^{n-1} v_{X_i} \quad (n \geq 1), \quad A_0 = 0,$$

and $Y_n := X_n^2 - A_n$. The quantity $A_\tau = \sum_{i=0}^{\tau-1} v_{X_i}$ represents the *total accumulated genetic variance* up to absorption.

(a) Show that $(Y_n)_{n \in \mathbb{N}}$ is a martingale with respect to $(\mathcal{F}_n)$.

(b) Deduce the value of $\mathbb{E}[A_\tau]$ as a function of x and N . (You may assume that $\mathbb{E}[\tau] < \infty$, which

will legitimize the passages to the limit; take care to justify them.)

Problem II — The Ballot Problem via a Reversed Martingale (20 pts)

(indicative time: 30 min)

Let $a, b \in \mathbb{N}^*$ be two integers with $a > b$. Set $n = a + b$. Two candidates A and B compete in an election: A receives exactly a votes and B exactly b votes. The n ballots are counted *one by one*, in a *uniformly random* order (all permutations of the ballots being equally likely).

For $k \in \{0, 1, \dots, n\}$, let L_k denote the lead of A after the first k ballots have been counted:

$$L_k = (\text{number of } A \text{ votes among the first } k) - (\text{number of } B \text{ votes among the first } k).$$

Thus $L_0 = 0$, $L_n = a - b > 0$, and $L_k - L_{k-1} \in \{+1, -1\}$ for all k . We want to compute

$$p = \mathbb{P}(L_k > 0 \text{ for all } k \in \{1, \dots, n\})$$

(the probability that A is *strictly in the lead throughout the entire count*).

Let $\xi_k \in \{+1, -1\}$ denote the contribution of the k -th ballot ($+1$ for A , -1 for B), so that $L_k = \sum_{i=1}^k \xi_i$. We introduce the *decreasing* filtration $(\mathcal{G}_k)_{0 \leq k \leq n}$ defined by

$$\mathcal{G}_k = \sigma(\xi_{k+1}, \xi_{k+2}, \dots, \xi_n), \quad k \in \{0, 1, \dots, n\},$$

(with $\mathcal{G}_n = \{\emptyset, \Omega\}$), so that $\mathcal{G}_0 \supseteq \mathcal{G}_1 \supseteq \dots \supseteq \mathcal{G}_n$. We set, for $k \in \{1, \dots, n\}$,

$$M_k = \frac{L_k}{k}.$$

(1) Verify that L_k is $\mathcal{G}_k$ -measurable. Then show that, for all $k \in \{1, \dots, n\}$,

$$\mathbb{E}[\mathbf{1}_{\{\xi_k = +1\}} \mid \mathcal{G}_k] = \frac{L_k + k}{2k}.$$

(2) Deduce that $(M_k)_{1 \leq k \leq n}$ is a *reversed martingale*, in the sense that, for all $k \in \{1, \dots, n-1\}$,

$$\mathbb{E}[M_k \mid \mathcal{G}_{k+1}] = M_{k+1}.$$

(3) Define the *last visit to 0*

$$R = \max \{ k \in \{1, \dots, n\} : L_k = 0 \},$$

with the convention $R = 1$ if this set is empty. Show that $\{R \leq k\} \in \mathcal{G}_k$ for all $k \in \{1, \dots, n\}$ (that is, R is a *stopping time for the reversed filtration*), and state precisely the version of the optional stopping theorem applicable to (M_k) and R , justifying its hypotheses.

(4) By applying optional stopping, show that $M_R = \mathbf{1}_{\{L_k > 0 \ \forall k\}}$ almost surely, and deduce that

$$p = \frac{a - b}{a + b}.$$

Problem III — Empty boxes, Doob martingale and Azuma–Hoeffding concentration

(21 pts)

(indicative time: 30 min)

We throw $n \geq 1$ balls independently into n boxes numbered $1, \dots, n$, each ball falling uniformly at random into one of the boxes. Formally, let $X_1, \dots, X_n$ be *i.i.d.* random variables, uniformly distributed on $\{1, \dots, n\}$, where X_i denotes the box of the i -th ball. We write

$$Z = \#\{j \in \{1, \dots, n\} : \text{box } j \text{ is empty}\} = \sum_{j=1}^n \mathbf{1}\{X_i \neq j \text{ for all } i \in \{1, \dots, n\}\}$$

for the number of boxes that remain empty. Set $\mathcal{F}_0 = \{\emptyset, \Omega\}$ and, for $1 \leq k \leq n$, $\mathcal{F}_k = \sigma(X_1, \dots, X_k)$.

Question 1. Compute $\mathbb{E}[Z]$ and show that $\mathbb{E}[Z] = n(1 - \frac{1}{n})^n$. Deduce that $\mathbb{E}[Z]/n \rightarrow e^{-1}$ as $n \rightarrow \infty$.

Question 2. For $0 \leq k \leq n$, set $M_k = \mathbb{E}[Z \mid \mathcal{F}_k]$.

(a) Justify that $(M_k)_{0 \leq k \leq n}$ is a martingale with respect to $(\mathcal{F}_k)$, with $M_0 = \mathbb{E}[Z]$ and $M_n = Z$.

(b) Show that $|M_k - M_{k-1}| \leq 1$ for all $1 \leq k \leq n$.

Question 3. We admit the Azuma–Hoeffding inequality: if $(M_k)_{0 \leq k \leq n}$ is a martingale with $|M_k - M_{k-1}| \leq c_k$ a.s., then for all $t \geq 0$,

$$\mathbb{P}(M_n - M_0 \geq t) \leq \exp\left(-\frac{t^2}{2 \sum_{k=1}^n c_k^2}\right).$$

(a) Deduce that for all $t \geq 0$,

$$\mathbb{P}(|Z - \mathbb{E}[Z]| \geq t) \leq 2e^{-t^2/(2n)}.$$

(b) Deduce that $Z/n \rightarrow e^{-1}$ in probability as $n \rightarrow \infty$. More precisely, for every $\varepsilon > 0$, bound $\mathbb{P}(|Z/n - e^{-1}| > \varepsilon)$ above by a quantity tending to 0, and specify the rate.

Problem IV — Poisson Galton–Watson process: extinction probability and the martingale $W_n = Z_n/\lambda^n$ (19 pts)

(indicative time: 28 min)

Let $\lambda > 0$ be a fixed parameter. We consider a *Galton–Watson process* (or Bienaymé–Galton–Watson process) $(Z_n)_{n \geq 0}$ with offspring distribution Poisson of parameter λ . Precisely, we are given an i.i.d. family $(\xi_{n,i})_{n \geq 0, i \geq 1}$ of random variables with law $\text{Poisson}(\lambda)$, that is,

$$\mathbb{P}(\xi = \ell) = e^{-\lambda} \frac{\lambda^\ell}{\ell!} \quad (\ell \in \mathbb{N}),$$

and we set $Z_0 = 1$ and then, for $n \geq 0$,

$$Z_{n+1} = \sum_{i=1}^{Z_n} \xi_{n,i} \quad (\text{empty sum} = 0 \text{ if } Z_n = 0).$$

We interpret Z_n as the size of the n -th generation, each individual of generation n independently giving birth to a $\text{Poisson}(\lambda)$ number of offspring. We denote by $\mathcal{F}_n = \sigma(Z_0, \dots, Z_n)$ the natural filtration. Recall that for a variable $\xi \sim \text{Poisson}(\lambda)$ we have $\mathbb{E}[\xi] = \lambda$ and $\text{Var}(\xi) = \lambda$, and its probability generating function is

$$f(s) = \mathbb{E}[s^\xi] = \sum_{\ell \geq 0} s^\ell e^{-\lambda} \frac{\lambda^\ell}{\ell!}, \quad s \in [0, 1].$$

Finally we denote by

$$q = \mathbb{P}(\exists n \geq 0 : Z_n = 0)$$

the *extinction probability* and by $f_n = f \circ f \circ \dots \circ f$ (n times) the n -th iterate of f .

Question 1 (generating function and mean).

(a) Compute $f(s)$ explicitly and show that $f(s) = e^{\lambda(s-1)}$ for all $s \in [0, 1]$.

(b) Recall (justifying it briefly from independence) that the generating function of Z_n is f_n , the n -th iterate of f . Deduce that $\mathbb{E}[Z_n] = \lambda^n$ for all $n \geq 0$.

Question 2 (extinction and criticality). We take for granted the result from the course: q is the smallest fixed point in $[0, 1]$ of the generating function f , that is, the smallest solution in $[0, 1]$ of the equation

$$q = f(q) = e^{\lambda(q-1)}.$$

- (a) Study the variations of f on $[0, 1]$ (convexity, values and slopes at the endpoints) and discuss, *according to the value of λ* , the number of fixed points of f in $[0, 1]$. Deduce that:

$$q = 1 \text{ if } \lambda \leq 1, \quad \text{and} \quad q < 1 \text{ if } \lambda > 1.$$

In particular, clarify the role of the *critical case* $\lambda = 1$ (where $f'(1) = 1$) and explain why extinction is almost sure there even though $\mathbb{E}[Z_n] = 1$ for all n .

- (b) For $\lambda = 2$, show that the extinction probability q is the unique solution in $(0, 1)$ of $q = e^{2(q-1)}$, and check that $q \in (0, 1)$ (no closed form is required; bound q , for example by showing $0.2 \leq q \leq 0.3$).

Question 3 (the martingale $W_n = Z_n/\lambda^n$). We set

$$W_n = \frac{Z_n}{\lambda^n} \quad (n \geq 0).$$

- (a) Show that $\mathbb{E}[Z_{n+1} \mid \mathcal{F}_n] = \lambda Z_n$ almost surely, then deduce that $(W_n)_{n \geq 0}$ is a positive martingale with respect to $(\mathcal{F}_n)$, and that $\mathbb{E}[W_n] = 1$ for all n .

- (b) Deduce that (W_n) converges almost surely to an integrable random variable $W_\infty \geq 0$ (state precisely the convergence theorem used and verify its hypotheses).
- (c) Assume we are in the *subcritical or critical* case $\lambda \leq 1$. Using Question 2, show that $W_\infty = 0$ almost surely. Deduce that, in this case, the convergence $W_n \rightarrow W_\infty$ *does not hold in* L^1 , and comment on what this reveals (failure of uniform integrability).

Problem V — Reflection Principle: Joint Law of the Maximum and the Hitting Time of Brownian Motion

(21 pts)

(indicative time: 30 min)

Let $(B_t)_{t \geq 0}$ be a standard Brownian motion started at 0 (that is, $B_0 = 0$, continuous paths, independent and stationary increments, $B_t \sim \mathcal{N}(0, t)$), constructed on a filtered probability space $(\Omega, \mathcal{F}, (\mathcal{F}_t)_{t \geq 0}, \mathbb{P})$ satisfying the usual conditions and for which (B_t) is strongly Markov. Fix $t > 0$ and write

$$M_t = \max_{0 \leq s \leq t} B_s.$$

For $a > 0$, set $\tau_a = \inf\{s \geq 0 : B_s = a\}$ (with the convention $\inf \emptyset = +\infty$).

Denote by $\varphi(x) = \frac{1}{\sqrt{2\pi}} e^{-x^2/2}$ the $\mathcal{N}(0, 1)$ density, by Φ its cumulative distribution function, and $\bar{\Phi} = 1 - \Phi$. Recall that B_t has a density (so $\mathbb{P}(B_t = c) = 0$ for every c).

(1) *Reflection principle.* Prove that for every $a > 0$,

$$\mathbb{P}(M_t \geq a) = 2 \mathbb{P}(B_t > a).$$

Deduce the density of M_t as well as the equality in law $M_t \stackrel{d}{=} |B_t|$.

(2) Establish, for every $a > 0$ and every $b \leq a$, the joint law in the form

$$\mathbb{P}(M_t \geq a, B_t \leq b) = \mathbb{P}(B_t \geq 2a - b).$$

Deduce that the pair (M_t, B_t) has, on the domain $\{a \geq 0, b \leq a\}$, the joint density

$$f_{(M_t, B_t)}(a, b) = \frac{2(2a - b)}{\sqrt{2\pi} t^{3/2}} \exp\left(-\frac{(2a - b)^2}{2t}\right).$$

(3) For $a > 0$, show that $\{\tau_a \leq t\} = \{M_t \geq a\}$, then deduce that τ_a has the density

$$f_{\tau_a}(s) = \frac{a}{\sqrt{2\pi} s^{3/2}} \exp\left(-\frac{a^2}{2s}\right), \quad s > 0.$$

Check that $\mathbb{P}(\tau_a < \infty) = 1$ and that $\mathbb{E}[\tau_a] = +\infty$.